

Same mixture, different mission

Learning intention

Choose and justify filtration, evaporation or evaporation with condensation for a stated recovery target.

Two teams receive saltwater.

One wants salt; one wants water.

Should their methods be identical?

Recover the mixture ideas

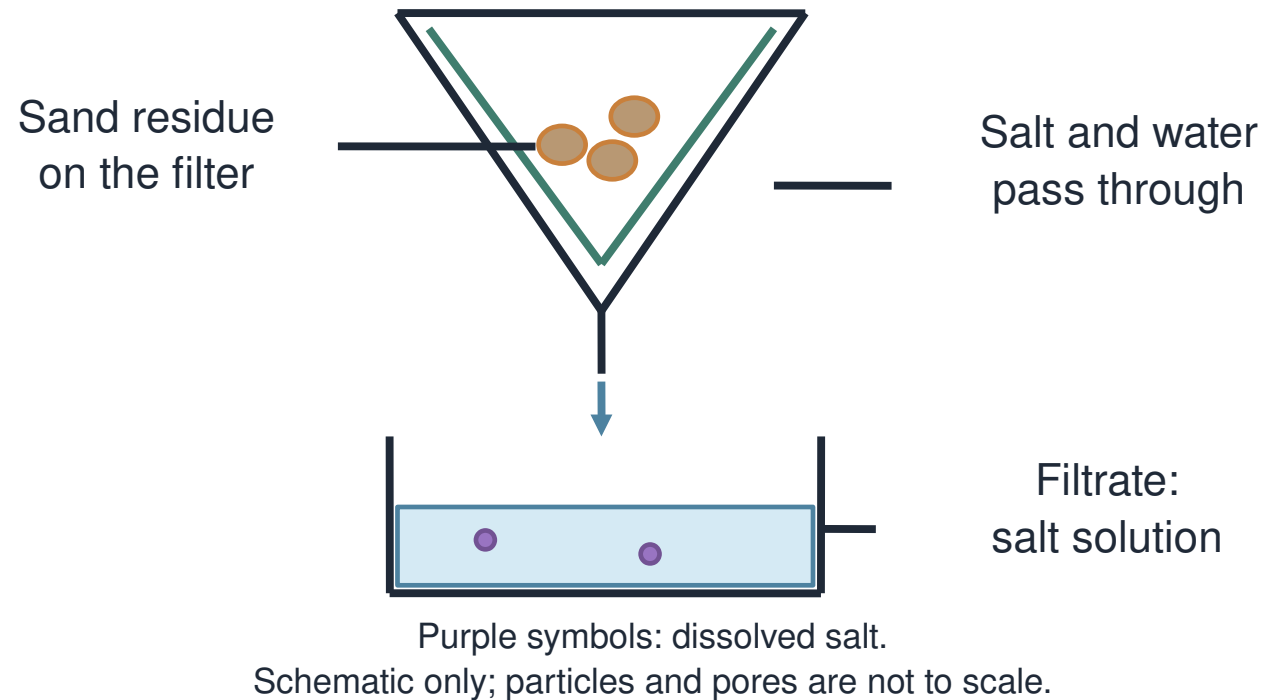
R1. Is dissolved salt still present?

R2. What does evaporation change?

R3. What is condensation?

I decide what the filter can retain

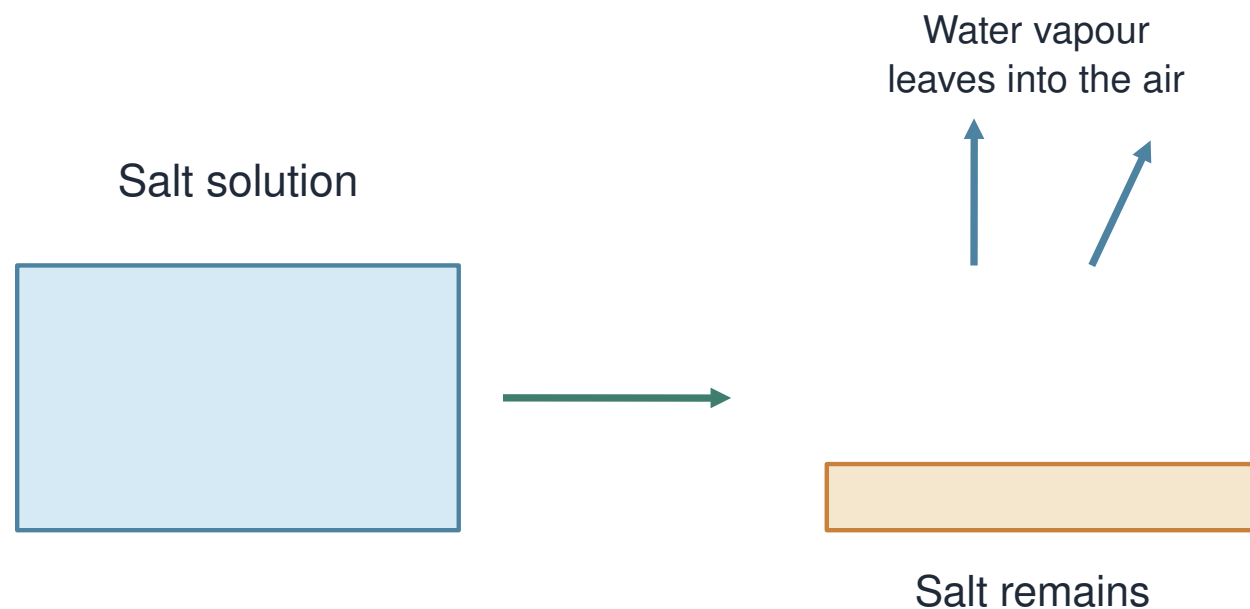
A filter retains sand, not dissolved salt



Sand remains as residue. Salt and water pass as filtrate. Clear filtrate is not proof of pure water.

I target the dissolved solid

Target: recover the salt

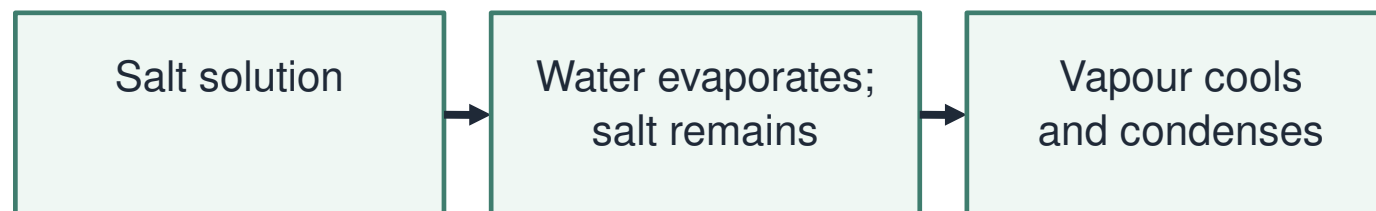


Open evaporation does not collect the water.

Removing water can leave salt behind. Open evaporation loses water into the air. This suits a salt target, not a collected-water target.

I change the plan when the target changes

Target: recover water from saltwater



Capture the vapour and collect the liquid water.
Process schematic only; not an apparatus-building guide.

To collect water, capture its vapour. Cool the vapour so it condenses. The collected liquid is the target.

Recovery workshop: choose, run, explain

W1. Choose methods for briefs A, B and C. Name the recovered substance each time. Test the process model or use paper arrows.

Brief	Mixture	Target
A	Sand + water	Sand
B	Saltwater	Salt
C	Saltwater	Water
D	Sand + saltwater	Sand and salt separately

Two targets need a sequence

W2. Plan brief D.

What must happen before salt recovery?

Identify residue and filtrate.

Repair the purity claim

W3. “The liquid passed a filter, so it is safe to drink.”

Use the dissolved-salt evidence to challenge it.

A team wants water from saltwater

Which process plan matches that target?

- A** Let water evaporate into the room and keep the salt.
- B** Capture water vapour and condense it into a collecting container.
- C** Filter once and assume all dissolved salt is trapped.

Your independent investigation

Choose one response route.

Match each mixture to its recovery target.

Explain where each component ends up.

Use the evidence

Name residue and filtrate when filtering.

Keep evaporation separate from collected condensation.

For more than one target, sequence the steps.

Check before sharing

Does my explanation answer the question?

Have I named the evidence or process?

Have I kept the conclusion within its limits?

Compare and improve

Trace one chosen route aloud or on paper.

Ask where the other component goes.

Improve a recovery or purity claim.

A question worth investigating

Suggest one next question about this lesson's evidence.

Choose a question whose answer would improve our model.

Show the science you can explain

E1. What does ordinary filtration remove here?

E2. How can salt be recovered from saltwater?

E3. What extra process collects the water?